



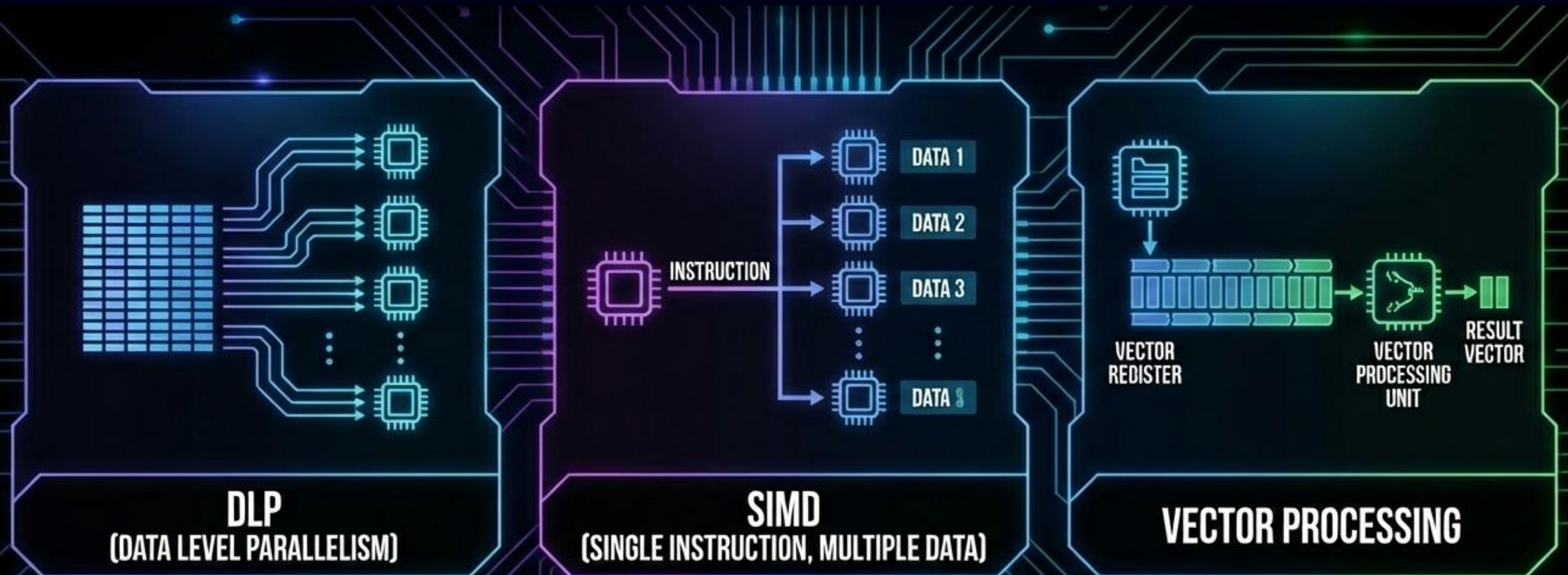
Data-Level Parallelism (DLP): SIMD & Vector Processing

Presentation Roadmap

- 1 Introduction to DLP, SIMD & Vector Processing
- 2 Working principle of SIMD & Vector Processing
- 3 Applications
- 4 Advantages & Disadvantages



Introduction to DLP, SIMD & Vector Processing



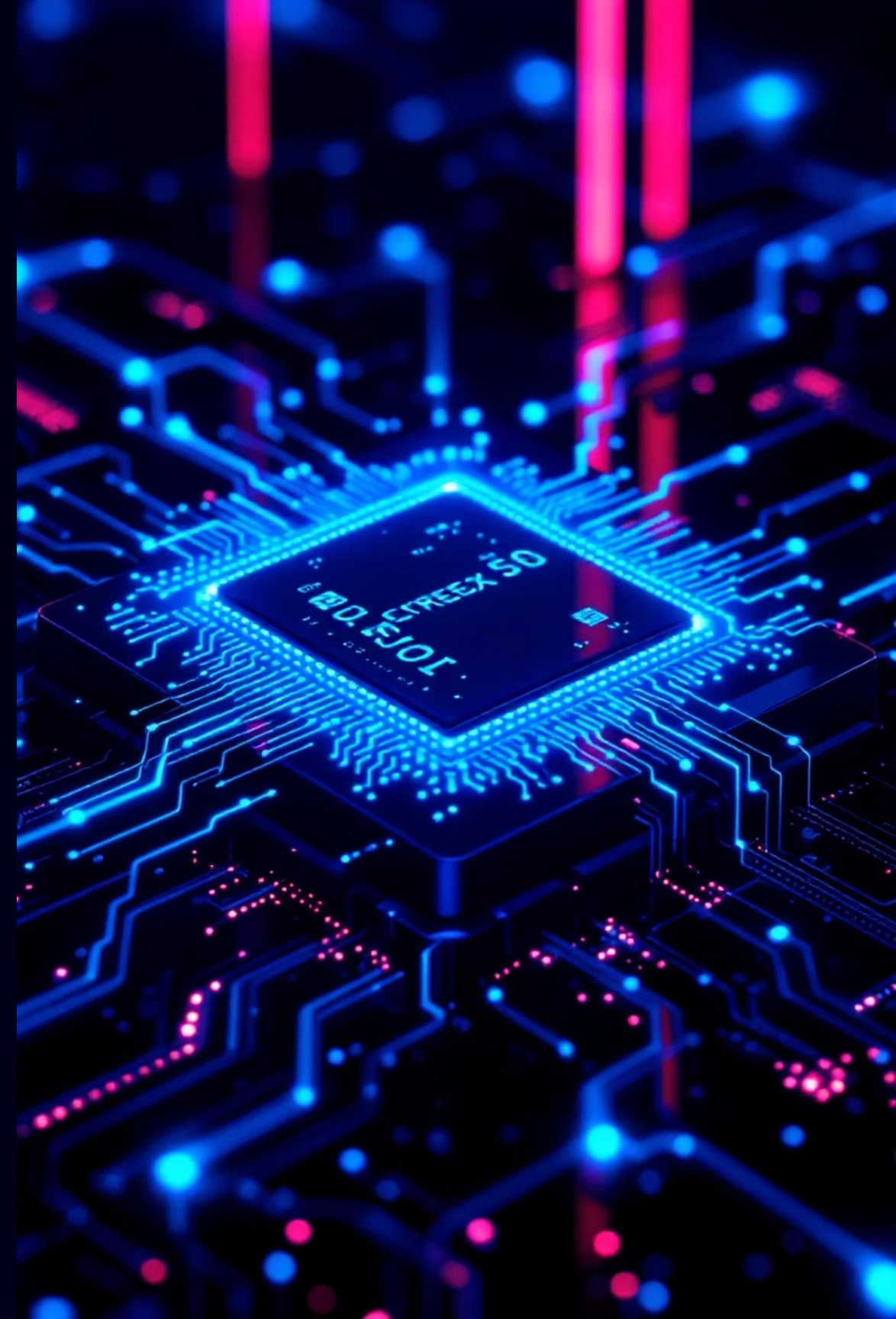
Parallel Processing

Multiple tasks

Executed at the same time



Data-Level Parallelism (DLP)



Data Level Parallelism (DLP)

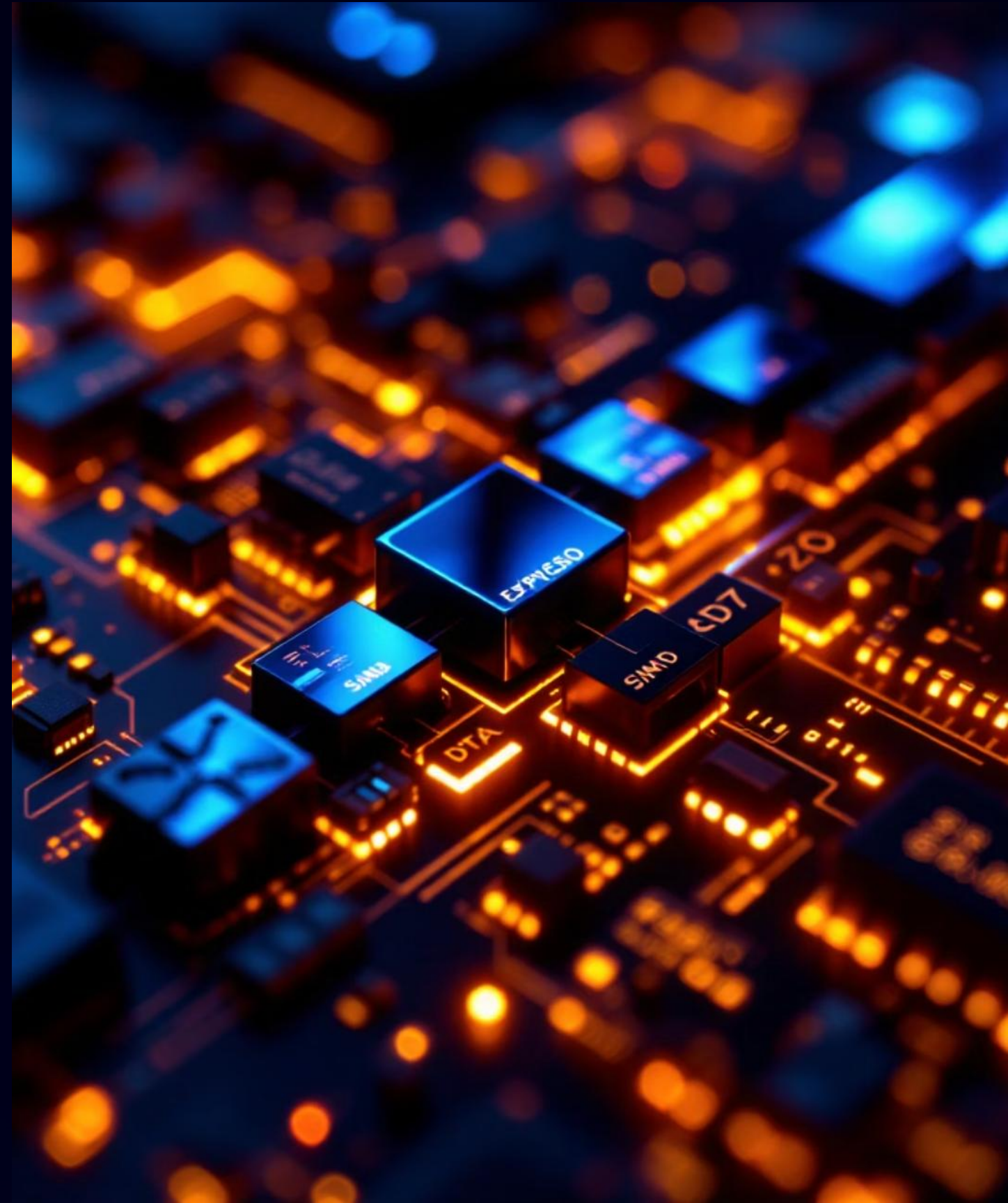


Same instruction

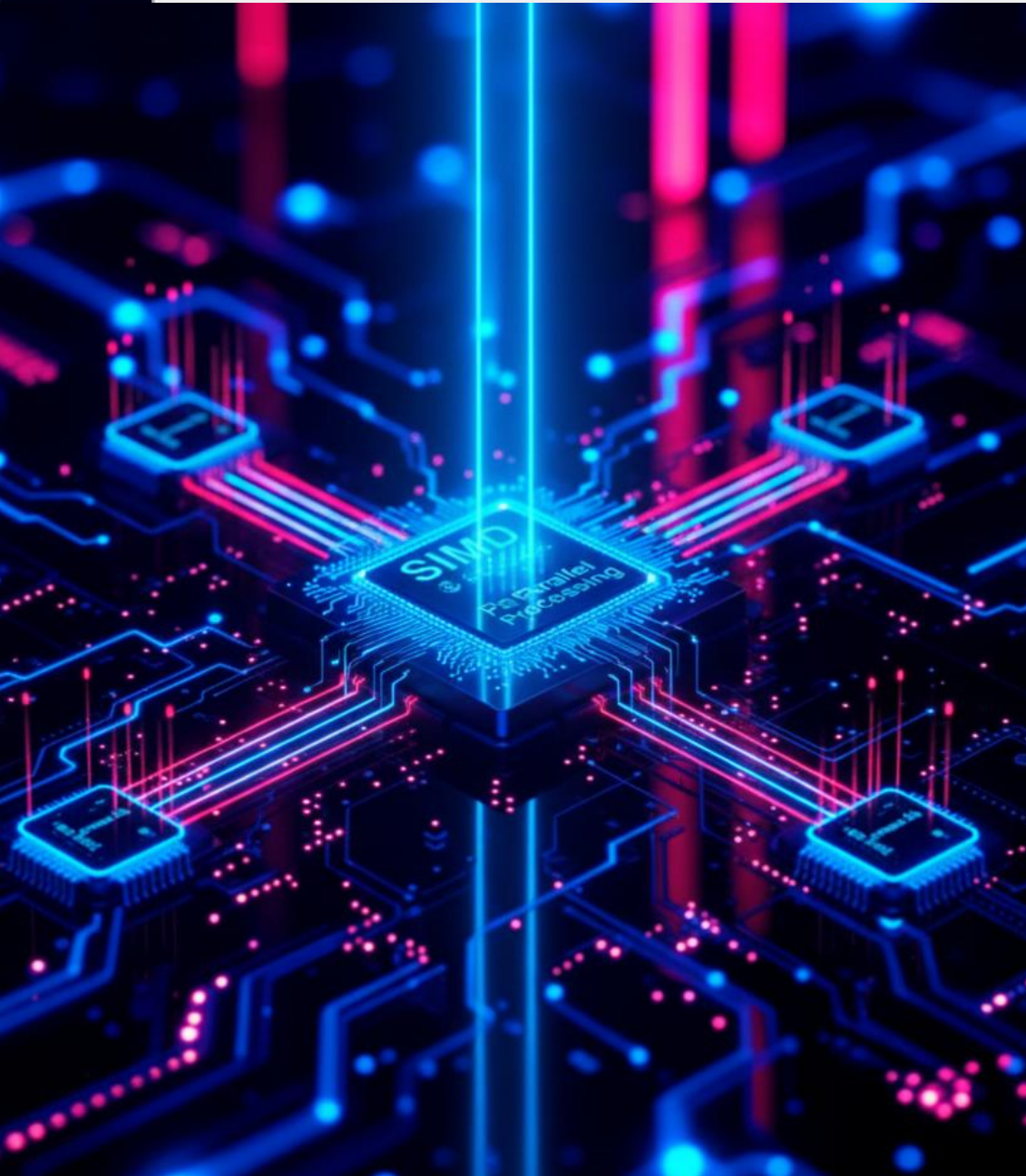
Operates on multiple
data elements

At the same time

SIMD (Single Instruction,
Multiple Data)



SIMD (Single Instruction, Multiple Data)



One instruction

Executed on multiple data
elements simultaneously

Example:

$A = [1, 2, 3, 4]$

$B = [5, 6, 7, 8]$

$A + B = [6, 8, 10, 12]$ (In one instruction)

Vector Processing: A Specialized SIMD Architecture



Vector Processing: A Specialized SIMD Architecture



Data is stored in vectors



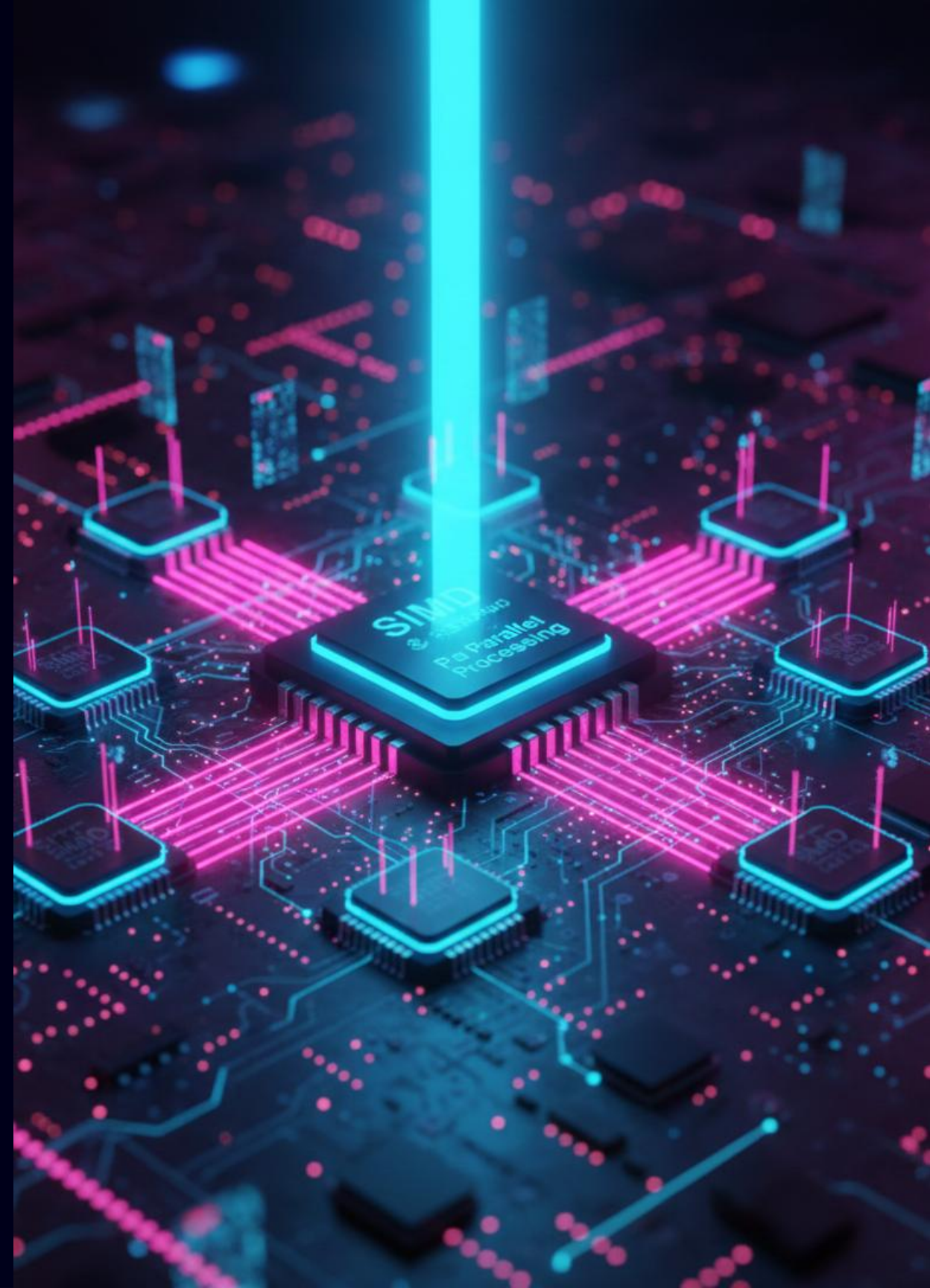
Operations are performed on entire vector



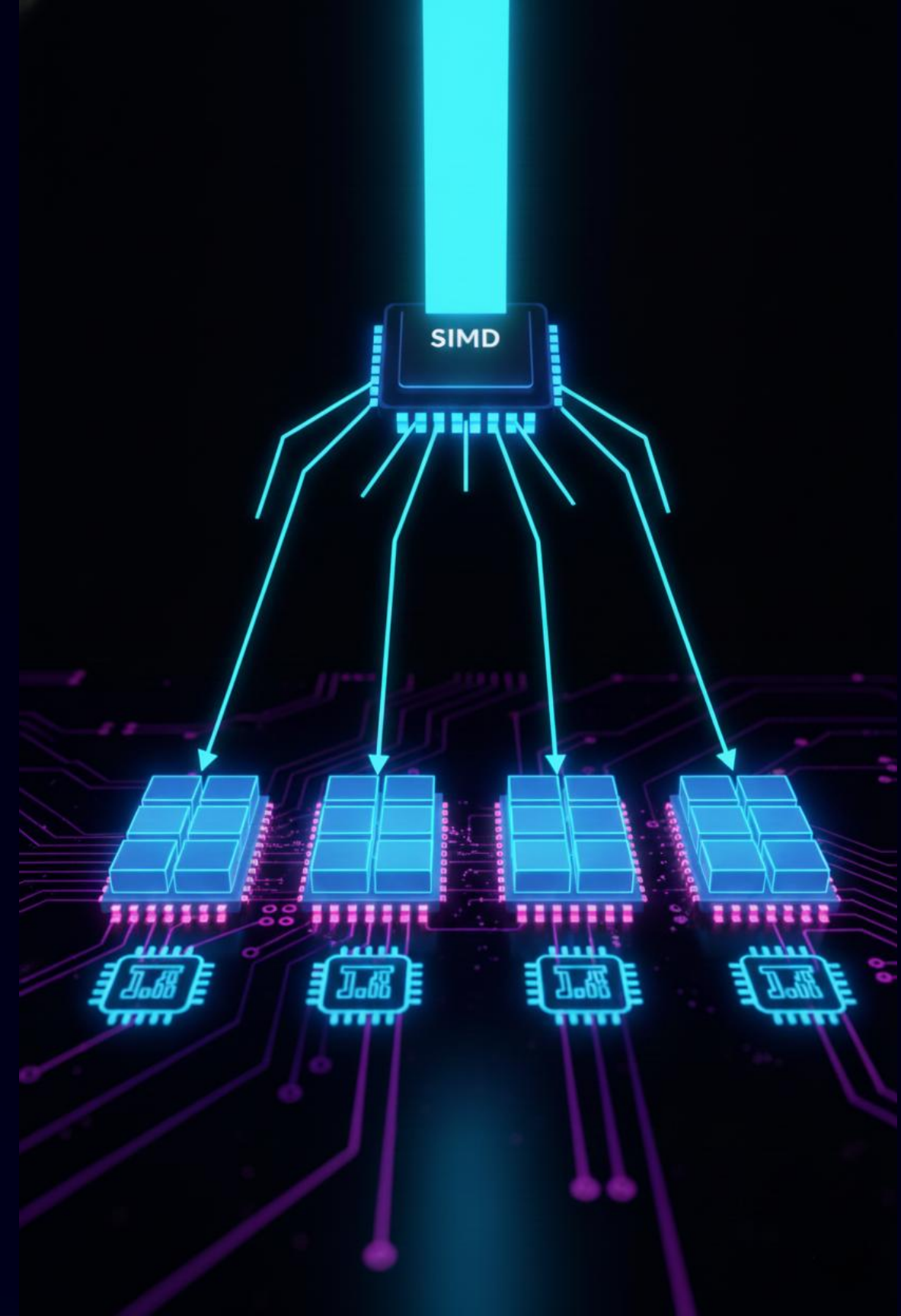
$$\begin{array}{c} \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 1 & 3 & 2 \end{pmatrix} \begin{pmatrix} 10 & 11 \\ 7 & 5 \\ 2 & 4 \end{pmatrix} \\ 3 \times 3 \quad 3 \times 2 \end{array} = \begin{array}{c} \begin{pmatrix} 1*10+2*7+3*2 & 1*11+2*5+3*4 \\ 4*10+5*7+6*2 & 4*11+5*5+6*4 \\ 1*10+3*7+2*2 & 1*11+3*5+2*4 \end{pmatrix} \\ 3 \times 2 \end{array}$$



Working Principle of SIMD & Vector Processing



SIMD Execution Inside the Processor



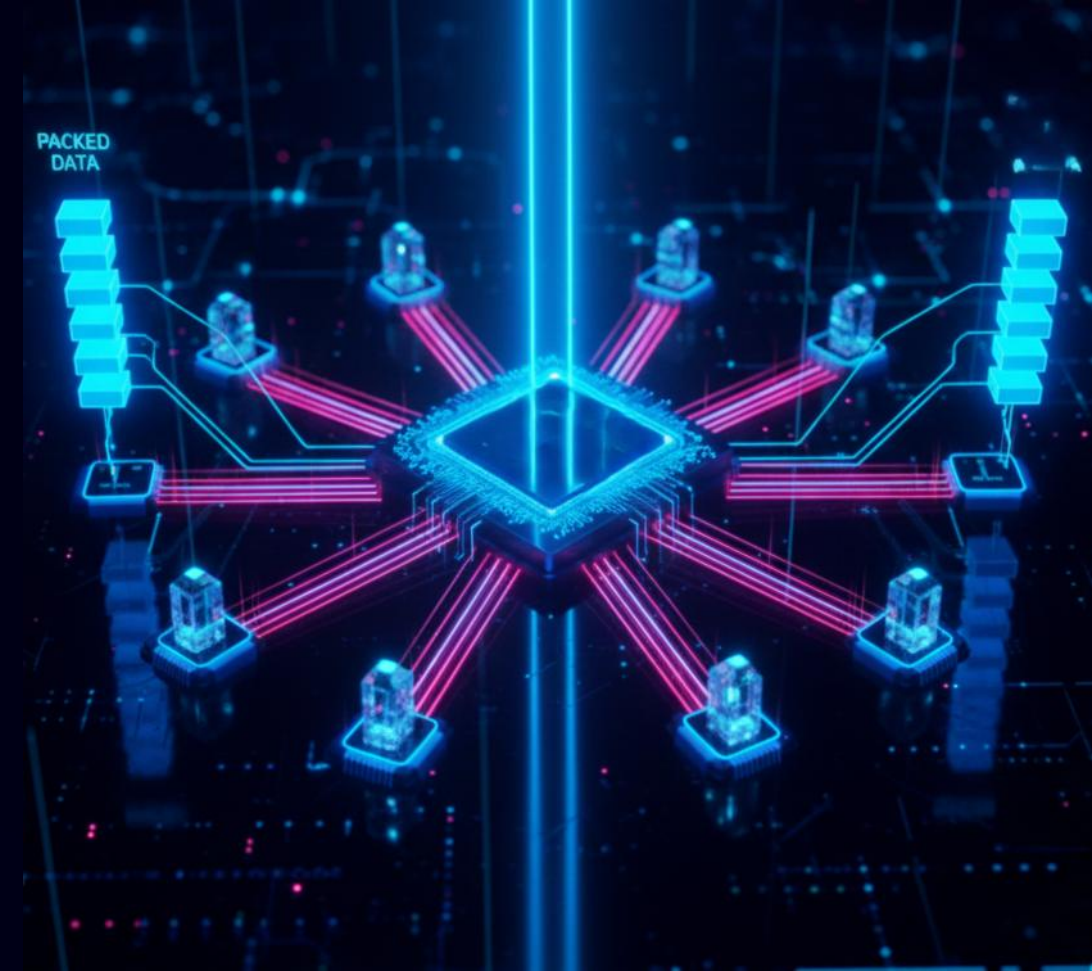
SIMD Execution Inside the Processor

Instruction Broadcast

- One instruction goes to all lanes
- Each lane works on one data element
- All lanes run in parallel

Packed Data

- Data stored in arrays or packs
- Elements are aligned in memory
- Proper alignment boosts efficiency

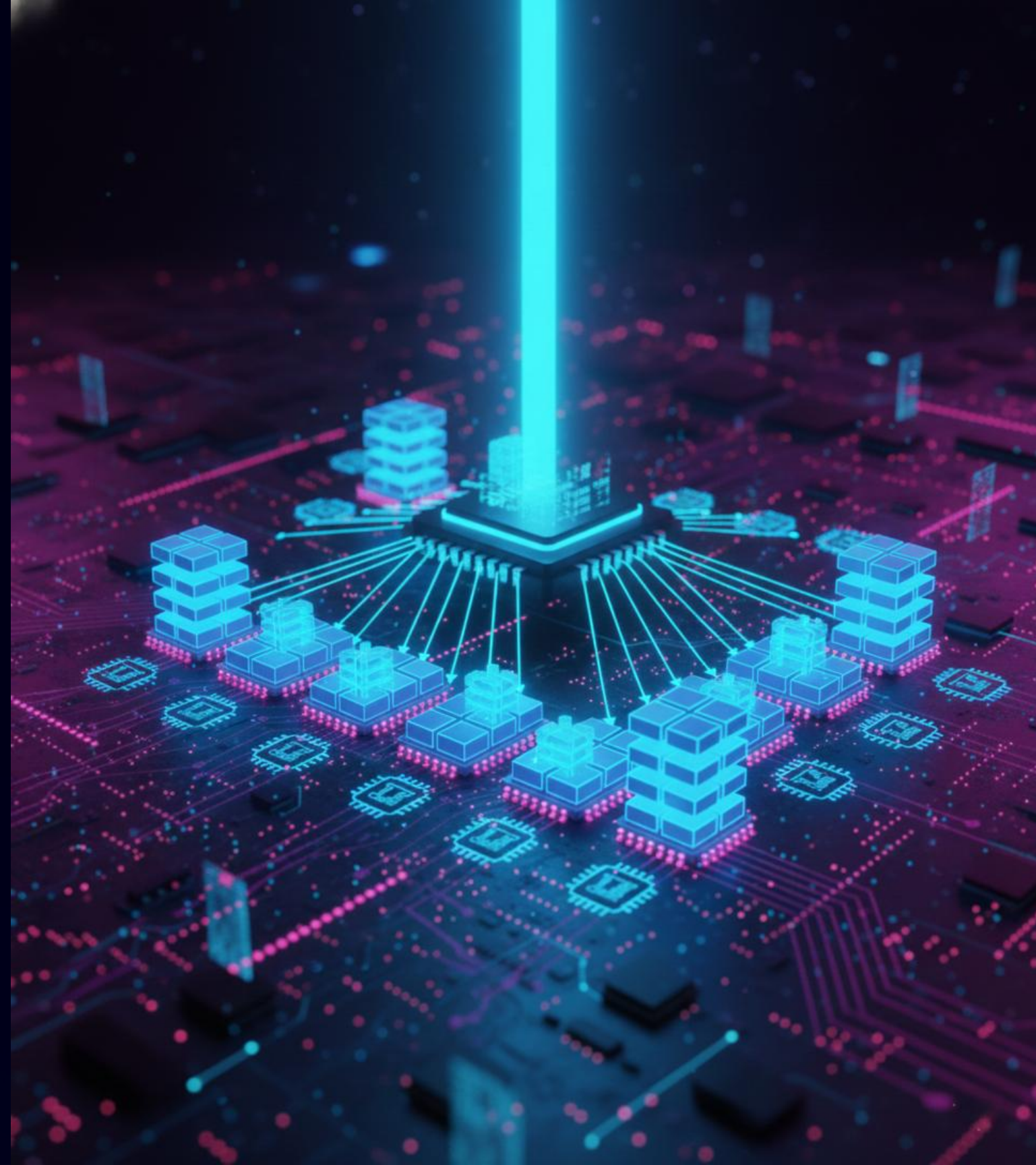


Instruction Broadcast

- Special instructions for vectors
- One instruction handles many values
- Fewer instructions save CPU time

Vector Processing

- **Advanced SIMD model**
- **Large data handling**
- **Specialized hardware**



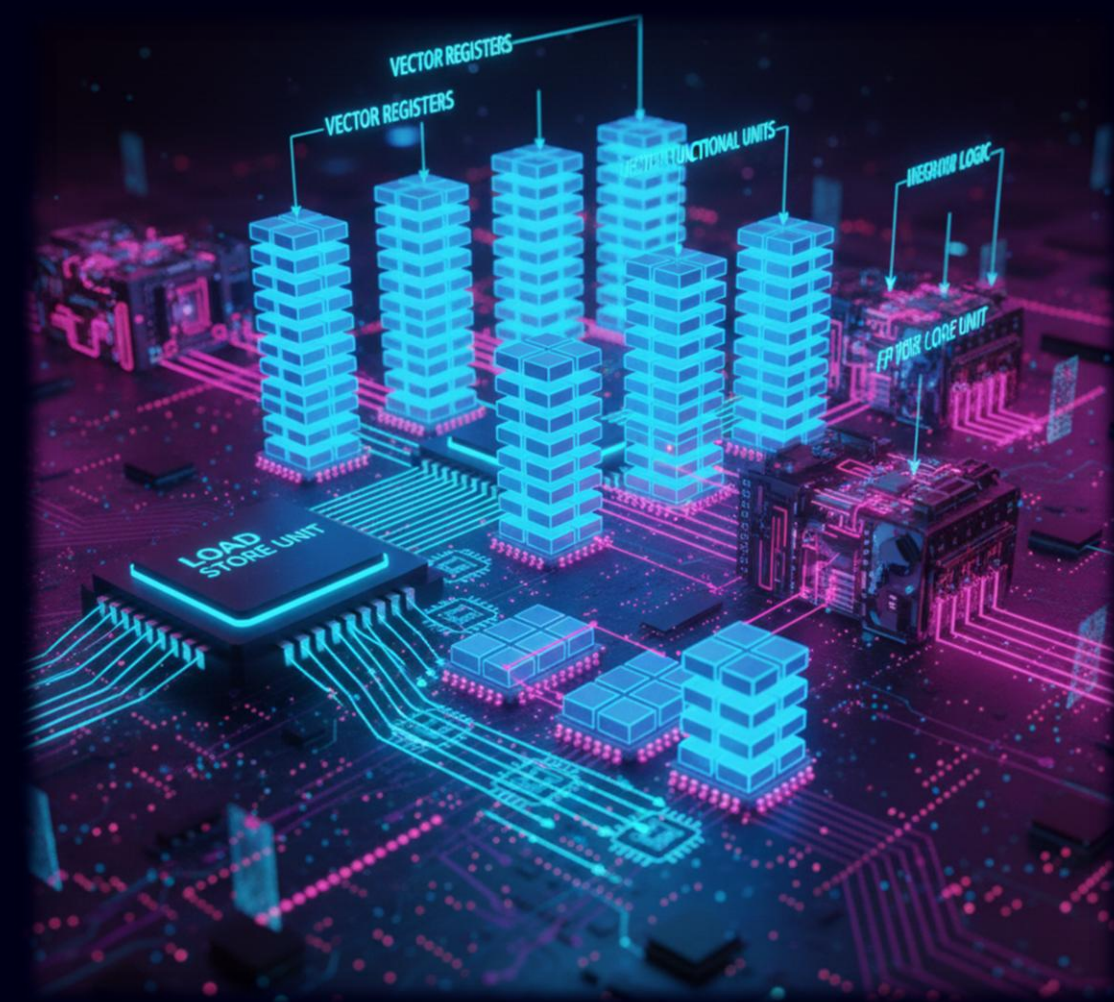
Vector Processor Components

Vector Registers

- Store many data elements at once
- Allow access to large block of data
- Enable fast, parallel processing

Vector Functional Units

- Perform operations on entire vectors
- Same operation applied to all elements
- Make computation fast and parallel



Load/Store Units

- Move data between memory and registers
- Optimized for high speed transfer
- Support efficient vector processing

Execution Flow

Load data



Execute vector operation



Store result



Advantages & Disadvantages of SIMD and Vector Processing



Advantages of SIMD and Vector Processing

- Faster Processing
- Saves Execution Time
- Better Use of CPU & GPU
- Energy Efficient
- Best for Large Data Set
- Improves Application Performance



Disadvantages of SIMD and Vector Processing

- Not Suitable for All Program
- Difficult Programming
- Less Flexible
- Hardware Dependent
- Wasted Resources
- Debugging is Hard



Conclusion





Conclusion

01

Performance Boost

DLP via SIMD and vector processing significantly enhances performance by parallelizing data operations.

02

Pervasive Technology

Integral to modern CPUs, GPUs, and various specialized processors.

03

Architectural Insight

Understanding these concepts is critical for designing efficient computer architectures.

Any Question ?

Thank You!